

Deep Learning Glossary

Activation: Function to make the output of a neuron non-linear (e.g., ReLU)

Architecture: the shape/definition of your model

Attention: mechanism telling the degree of importance each feature should pay to each other (self-attention)

Auto-Encoder: a type of architecture that compresses and decompresses the data itself, usually in an unsupervised manner.

Batch: subdivision of your training data to train with randomness

Convergence: when the model does not learn anymore

Convolution: a mathematical operation where you slide a kernel/filter along your data, transforming each part in the same manner.

Dense layer: layer using classic/fully connected neurons (not convolution)

DL: Deep Learning

Dropout: a method where you set to 0 some weights during training. Help preventing overfitting

Embedding: an intermediate representation of your data learned by your model

Epoch: each time the model has seen all your training data

Features: the input of your model (e.g. “genes”, “pixels”); sometimes used interchangeably with embeddings.

Filter: matrix used to transform an image, e.g. by convolving it with the image. Synonym of Kernel. It can be seen as the convolutional counterpart to a neuron in a dense layer.

Foundation model: a model trained with a great amount of data, learning embeddings that are supposed to be a general representation of the input. Usually trained with self- or un- supervised learning.

Fully Connected layer: Same as Dense layer

Gradient: derivative in multi-dimension; used to find in which direction you update the weights (gradient descent).

Ground truth: the labels Y that you already know in your training set

Kernel: same as filter

Key (transformers): representation of a feature which represents what about it can be useful for other. By multiplying it by all other Queries, you learn attention.

Latent Space: optimised / Compressed space where from where our data can be generated.

Layer: a vertical stack of neurons; the input of a layer is the output of the previous layer.

LLM: Large Language Model

Loss: the objective functions to define what **error** your models perform

Model: the function you use to predict labels from input features (usually noted “f”)

MSE: Mean Square Error: simpler loss function

Neuron: the basic brick of deep learning algorithms

Optimisation: updating of the model's parameters to make better predictions / lower loss

Over-fitting: when your model learns by heart your data, failing to generalise to unseen data

Padding: adding 0 to the edges of your input so that your convolution can create an output of the same shape as your input.

Pooling: method used to compress data: e.g., max-pooling: every 4 consecutive pixels are replaced by the max pixel. Same with mean pooling.

Predictions: the output of your model

Query (transformers): representation of a feature which represents what would be useful to learn from others. By multiplying it by all other Keys, you learn attention.

Self-supervised learning: a type of learning where you can create “dummy” labels Y from X itself.

SGD: stochastic gradient descent: method to optimise (with batches for stochasticity)

Step: each time you update your model with gradient descent

Stride: how much your convolutional kernel “jumps” during the convolution. A stride of 2 will produce an output 2 times smaller (squared if 2D)

Test set: the data you use to test your model after training: never train on it!

Training set: the data you used for training

Transformer: architecture using attention and dense layers to create embeddings aware of and informed by their context.

Translation invariance: when your signal does not change when you move it somewhere else in your data (by translation)

Unsupervised learning: A type of training where you only need your input data (X) and not any labels (Y)

Validation set: the data you used to check if the model is overfitting: you compute the loss but don't optimise on it!

Value (transformers): the embedding that is modified through attention.

Weights: the **parameters** of your model that you are updating to learn

X: usually your input data

Y: usually your labels / what you want to predict